CGS402: Assignment 1

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Introduction

This report presents findings from a human decision making study conducted with 24 participants. Each participant completed one of two forms containing 6 questions regarding various behavioral tasks designed to test well-known cognitive biases and heuristics, namely - Anchoring Effect, Framing Effect, Representativeness Heuristic, Choice Overload, Sunk Cost Fallacy and Status Quo Bias.

The links to the Google Forms created for this assignment can be found here:

- **Form A**: `https://forms.gle/3qb1h6JBgMP5Pxjw9`
- **Form B**: `https://forms.gle/X9iX6qoMpFJZhPHeA`

1 Heuristics & Biases

1.1 Anchoring Effect

1.1.1 What is the Anchoring Effect?

The anchoring bias is a cognitive bias that causes us to rely heavily on the first piece of information we are given about a topic. When making estimates or decisions under uncertainty, we interpret newer information from the reference point of this initial "anchor" instead of evaluating it objectively, and our subsequent adjustments away from the anchor tend to be insufficient.

1.1.2 Method

Participants were randomly assigned to one of two conditions regarding population of Argentina:

- **Low anchor group** (n=12): Asked "Do you think the population of Argentina is more or less than **1 crore**?"
- **High anchor group** (n=12): Asked "Do you think the population of Argentina is more or less than **10 crores**?"

After responding to the anchor question, all participants were asked to estimate the actual population of Argentina in crores.

1.1.3 Results

- **Low anchor group**: N = 12, Estimated Mean = 2.62 crores, SD = 2.17
- **High anchor group**: N = 12, Estimated Mean = 9.08 crores, SD = 8.86
- **Difference**: 6.47 crores (high anchor group estimated 3.47× higher)
- In the **low anchor group**, those who said "More" estimated a mean of 3.62 crores — (2.62 crores above the anchor of 1 crore). Those who said "Less" estimated a mean of 0.60 crores — (only 0.40 below the anchor), showing extreme anchoring.
- In the **high anchor group**, the 10 participants who said "Less" estimated a mean of 5.40 crores — (only 4.60 below the anchor of 10) while the 2 participants who said "More" estimated a mean of 27.50 crores — (17.50 above the anchor). The numbers are still very close to the anchor of 10 crores.

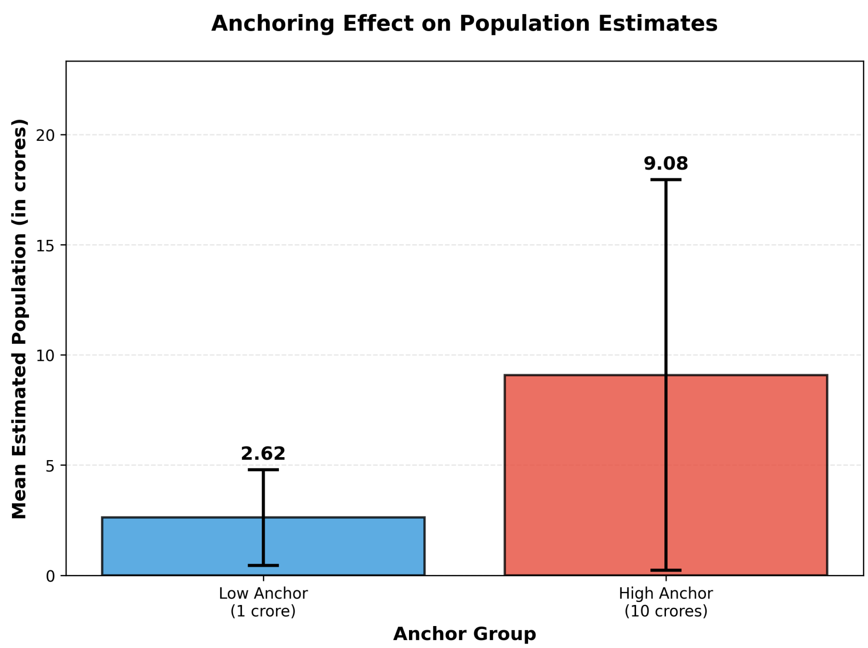

Figure 1: Mean estimated population of Argentina by anchor condition.

1.1.4 Alignment with Theory

This pattern is consistent with the **anchoring heuristic**, where people use an initial value as a starting point and make insufficient adjustments from it when forming estimates. Some possible explanations for this are:

1. **Insufficient adjustment**: People of study might not have much information about a country like Argentina. So, when faced with such an uncertain quantity, people start from the provided anchor and mentally adjust upward or downward. However, these adjustments are typically insufficient — people stop adjusting once they reach the first plausible value. In our data, the low anchor group adjusted upward from 1 crore but stopped at mean 2.62 crores, while the high anchor group adjusted downward from 10 crores but only reached 9.08.

2. **Selective accessibility**: After seeing the anchor, people retrieve information from memory that is consistent with it. E.g, when participants are told "1 crore," they think why the population might be small (e.g., Argentina is not well-known, South American countries seem smaller), while when told "10 crores," they retrieve information consistent with a large population (e.g., Argentina is a geographically large country). These reasonings lead them to make estimates hooked to the anchor.

1.2 Framing Effect

1.2.1 What is the Framing Effect?

The framing effect is a cognitive bias in which our decisions are influenced by the way information is presented, rather than by the information itself. Equivalent information can appear more or less attractive depending on whether its positive or negative attributes are highlighted.

1.2.2 Method

Participants were randomly assigned to one of two framing conditions describing an identical medical treatment outcome:

- **Gain frame** (n=12): "A new medical treatment for a dangerous virus **saves 200 out of 600 patients**. Would you recommend it?"
- **Loss frame** (n=12): "A new medical treatment for a dangerous virus results in **400 out of 600 patients dying**. Would you recommend it?"

1.2.3 Results

- **Gain frame**: N = 12, Estimated Mean = 4.42, SD = 1.56
- **Loss frame**: N = 12, Estimated Mean = 2.92, SD = 1.51
- **Difference:** 1.5 rating points higher for gain frame
- The gain frame mean (4.42) is above the midpoint 4, indicating a positive recommendation tendency.
- The loss frame mean (2.92) is below the midpoint 4, indicating reluctance to recommend.

1.2.4 Alignment with Theory

The results provide strong support for the framing effect. Participants' recommendations were significantly influenced by whether the outcome was presented in terms of lives saved (gain) or lives lost (loss), despite the statistical equivalence of both descriptions. Some possible explanations for this are:

1. **Loss Aversion Tendency**: The phrase "400 die" triggers a stronger negative emotional reaction while "200 saved" triggers a positive one, because the brain processes losses and negative information with greater intensity (survivalship bias).
2. **Lack of Mathematical Evaluation**: Rather than computing the objective survival rate ($\frac{200}{600}$ = 33.3%), participants substituted the easier question "Does this *feel* appropriate or not?" — and the answer to that easier question depends heavily on which frame is presented.

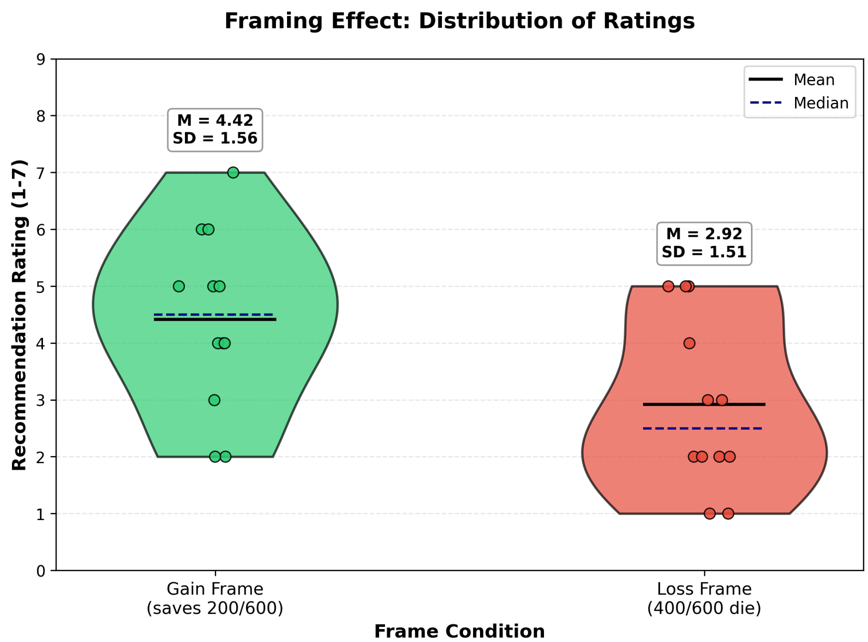

Figure 2: Mean recommendation ratings by frame condition.

This demonstrates the power of linguistic framing in decision-making and suggests that people do not evaluate outcomes in an objective, invariant manner, rather how they *feel* about it.

1.3 Representativeness Heuristic (Linda Problem)

1.3.1 What is the Representativeness Heuristic?

The representativeness heuristic is a mental shortcut we use when estimating probabilities. Instead of relying on logic, we judge how likely something is by assessing how similar it is to an existing mental prototype or stereotype. This can lead to errors such as the **conjunction fallacy** — the mistaken belief that a specific, detailed scenario is more probable than a broader one — because the detailed version "fits" our mental image better.

1.3.2 Method

All 24 participants were presented with the following description:

> "Rekha is 31, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice."

Participants were then asked to choose which statement is more probable:

- **(a)** Rekha is a bank teller
- **(b)** Rekha is a bank teller AND is active in the feminist movement

Logically, option (a) must be more probable than option (b), since the probability of conjunction of two events cannot be more probable than either event alone. However, the description is designed to make Rekha seem representative of a feminist, potentially leading participants to commit the conjunction fallacy.

1.3.3 Results

- **Option (a) - Bank teller only**: 4 participants (16.7%)
- **Option (b) - Bank teller AND feminist**: 20 participants (83.3%)
- **Conjunction fallacy rate**: 83.3% (some participants who marked correctly admitted having prior exposure to the Linda problem)

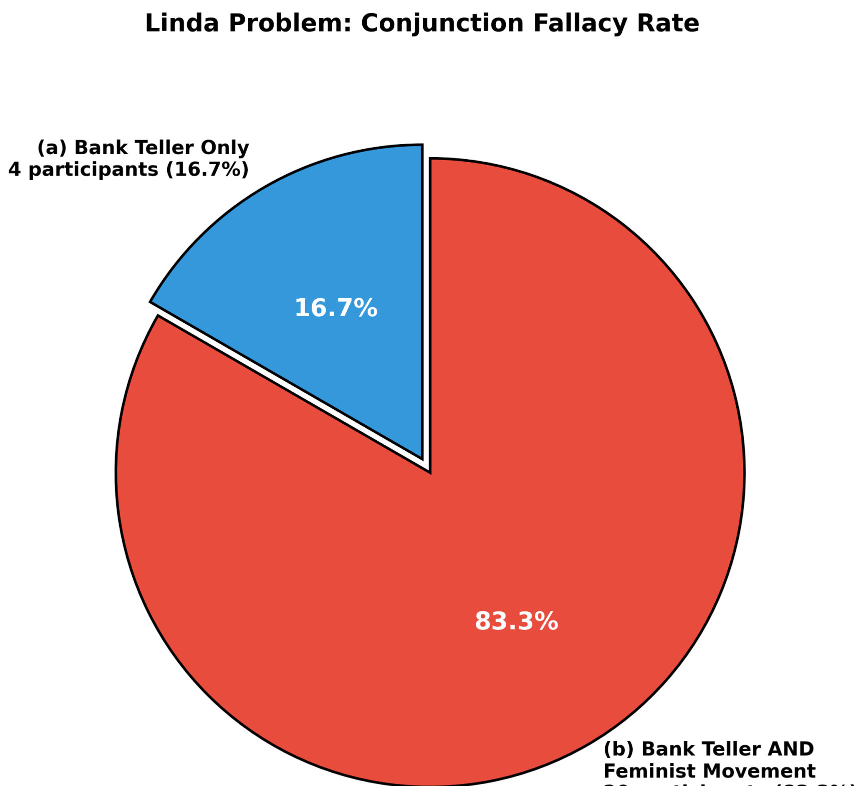

Figure 3: Distribution of responses to the Linda problem.

1.3.4 Alignment with Theory

This finding aligns with the representativeness heuristic, which says that people judge probability based on how well an instance matches a prototype or stereotype, rather than on logical probability principles. Some possible explanations for this are:

- **Prototype matching**: Rekha's description (philosophy major, concerned with discrimination and social justice, outspoken) closely matches the cultural prototype of a feminist activist. When asked about probability, participants substitute the question "How probable is this?" with the easier question "How well does Rekha *resemble* a feminist?" — and the answer to the latter is "very well."

- **Base Rate Neglect**: The set of "bank tellers" necessarily includes all "bank tellers who are feminists" as a subset. However, participants neglect this logical structure because the representativeness signal is so strong that it overwhelms formal reasoning.

2 Decision Making

2.1 Choice Overload

2.1.1 What is Choice Overload?

Choice overload describes how people get overwhelmed when they are presented with too many options. While we tend to assume that more choice is a good thing, research has shown that in many cases, having a larger array of options makes it harder to decide, reduces satisfaction with the chosen option, and can even lead people to defer the decision entirely.

2.1.2 Method

All 24 participants were presented with a hypothetical jam-buying scenario and rated their purchase likelihood in two conditions:

- **6 varieties condition**: "Imagine you walk into a store to buy jam. There are 6 varieties available. How likely would you be to actually buy a jar? (Rate 1-7)"
- **24 varieties condition**: "Now imagine there are 24 varieties available. How likely would you be to actually buy a jar? (Rate 1-7)"

2.1.3 Results

- **6 varieties**: N = 24, Estimated Mean = 4.79, SD = 1.86
- **24 varieties**: N = 24, Estimated Mean = 5.04, SD = 1.97
- **Difference**: -0.25 points (24 varieties condition mean slightly higher)
- The difference of 0.25 points is small and not statistically significant ($p = 0.601$).
- Both conditions show moderate purchase likelihood (means near the midpoint of 4).
- Standard deviations are similar (1.86 vs 1.97), indicating comparable variability in both.
- Individual ratings show considerable variation in both conditions.

2.1.4 Alignment with Theory

The results do **not support** the choice overload hypothesis. Participants actually reported slightly higher purchase likelihood when presented with 24 varieties compared to 6 varieties, though this difference was not statistically significant. Some possible explanations for this are:

- **Hypothetical vs. real choice**: Our study is a hypothetical environment rather than actual purchase behavior. So people may not experience the same cognitive burden as in real decision-making contexts. People may find variety to be more appealing rather than overwhelming when thinking abstractly.

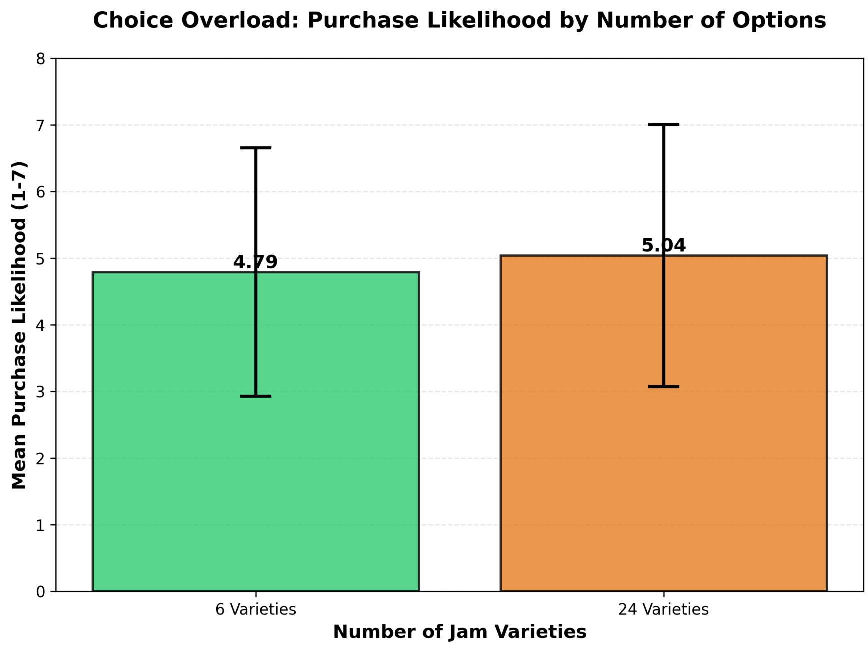

Figure 4: Mean purchase likelihood by number of jam varieties.

- **Sample size**: With only 24 participants, the study may lack power to detect small effects.
- **Satisfaction Test**: The question asks "likelihood" of buying in the two cases rather than "satisfaction" of buying in the two cases, failing to appropriately capture choice overload.

2.2 Sunk Cost Fallacy

2.2.1 What is the Sunk Cost Fallacy?

The sunk cost fallacy is our tendency to continue pursuing an endeavor once we have already invested time, money, or effort into it, even when giving up would be the more rational choice. Because sunk costs are irrecoverable regardless of what we decide next, they should logically be irrelevant to current decisions. However, the emotional pain of "wasting" a prior investment and a desire for self-justification leads us to sticking.

2.2.2 Method

All 24 participants were presented with two movie ticket scenarios:

- **Paid ticket condition**: "You've bought a movie ticket for 500. Thirty minutes in, you're really not enjoying the film. Would you stay or leave?"
- **Free ticket condition**: "Now imagine the ticket was free. Same bad movie. Would you stay or leave?"

Rationally, the decision should be the same in both cases—the money is already spent (sunk cost) and cannot be recovered, so the only relevant factor is whether continuing to watch is worth the time. Time is the resource at stake now. However, the sunk cost fallacy predicts that people will be more likely to stay when they have paid, irrationally trying to "get their money's worth."

2.2.3 Results

- **Paid ticket**: N=24, Stay = 23 (95.8%), Leave = 1 (4.2%)
- **Free ticket**: N=24, Stay = 7 (29.2%), Leave = 17 (70.8%)
- **Difference**: 66.7 percentage points increase in staying when paid.

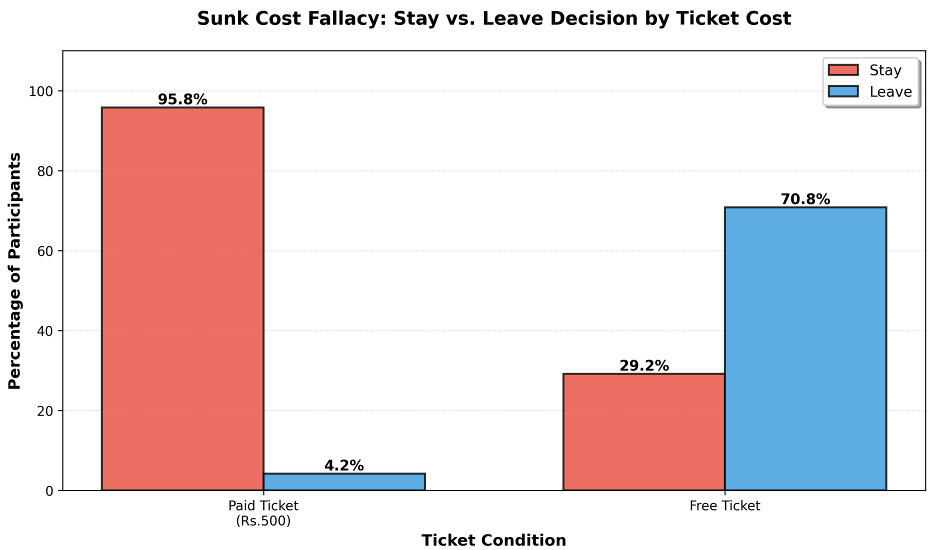

Figure 5: Stay vs. Leave decisions by ticket cost condition.

2.2.4 Alignment with Theory

This finding aligns perfectly with **sunk cost fallacy theory** which states people try to avoid "wasting" their prior expenditure. The 500 payment creates a psychological commitment that makes people reluctant to leave, even though the money cannot be recovered whether they stay or leave. Some possible explanations for this are:

- **Loss aversion**: Leaving feels like accepting a loss of 500. They want to feel they "got their money's worth," even if it means enduring a bad experience.
- **Self-justification**: Having made the investment creates pressure to justify the decision by completing the activity. The 500 is mentally "allocated" to watching the movie, making it hard to abandon.

2.3 Status Quo Bias

2.3.1 What is the Status Quo Bias?

The status quo bias describes our preference for the current state of things, resulting in resistance to change. When faced with a decision, we tend to favor the default or existing option over alternatives, even when switching could be beneficial.

2.3.2 Method

All 24 participants were presented with a phone plan switching scenario:

> "Your phone plan currently gives you 10 GB data and 200 minutes of calls. A new plan offers 15 GB data and 100 minutes of calls for the same price. Would you switch?"

Participants rated their likelihood to switch on a 7-point scale (1 = Definitely not, 7 = Definitely yes). The new plan offers more data but fewer call minutes at the same price—a trade-off that requires evaluation. Status quo bias would predict reluctance to switch despite the new plan potentially being better for many users (given modern usage patterns favor data over calls).

2.3.3 Results

The analysis revealed mixed evidence for status quo bias (N=24):

- **Mean rating**: 4.25 (SD = 2.19)
- **Median**: 4.0

Distribution by category:

- **Low switching willingness (1-2)**: 7 participants (29.2%)
- **Neutral (3-5)**: 8 participants (33.3%)
- **High switching willingness (6-7)**: 9 participants (37.5%)

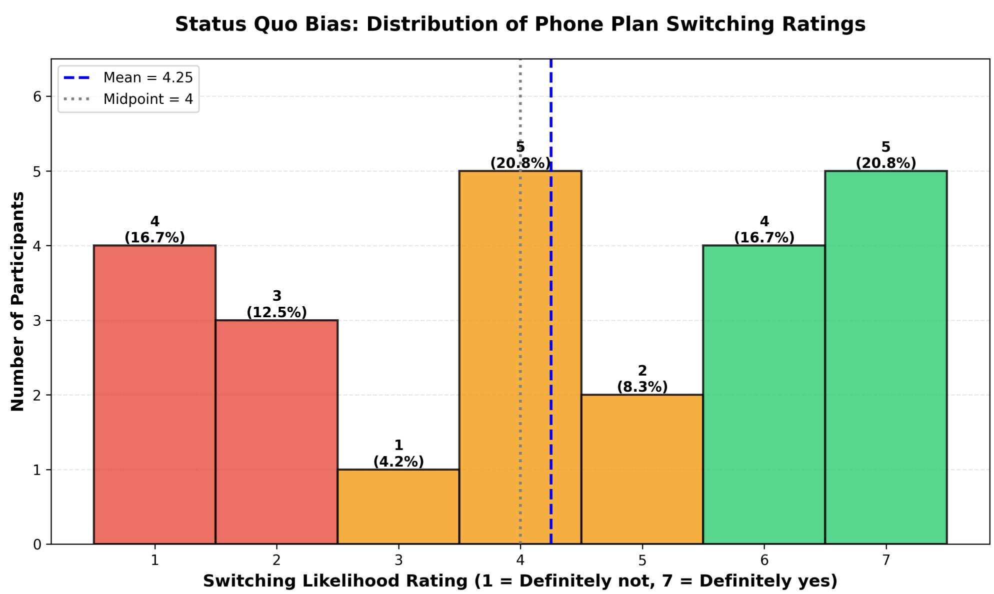

Figure 6: Distribution of phone plan switching likelihood ratings.

- The mean rating (4.25) is only slightly above the midpoint (4), indicating neither strong reluctance nor strong willingness to switch.
- High variability (SD = 2.19) suggests individual differences in how people evaluate the trade-off.

2.3.4 Alignment with Theory

The results show **weak or no evidence** for status quo bias in this context. Rather than showing a clear preference for maintaining their current plan, participants were relatively evenly distributed, with a slight tendency toward being willing to switch. Some of the possible explanations for this are:

- **Clear advantage**: The new plan offers 50% more data (15 GB vs. 10 GB) with only a 50% reduction in calls (100 vs. 200 minutes). For many modern users who rely heavily on data and rarely use voice calls using SIM, this is an obvious improvement. This is because nowadays we also have high prevalence of calls using data making it even more desirable, potentially overriding status quo bias.

- **Individual differences**: The high variability suggests that responses depend on individual usage patterns—heavy call users would resist switching, while data-heavy users would embrace it. Also, the scenario is hypothetical without real consequences, potentially reducing the psychological attachment to the "current" plan.

Status quo bias is more context-dependent rather than absolute. It may emerge more strongly when choices are complex, ambiguous, or involve switching costs—conditions not present in our scenario.

Conclusion

Our study confirmed that cognitive biases are not abstract concepts—they reliably shape how people think, estimate, and decide. Four of six biases (anchoring, framing, representativeness, and sunk cost) manifested strongly, while choice overload and status quo bias were weaker in our sample.

Since we are now better informed about these shortcuts that we take, we are more capable of avoiding them. The most effective defence is rethinking - structured comparisons, and deliberate reframing that uncovers the real aspects of situations and helps us make better judgements, which is a key to good decision-making.

References

1. Lectures slides of the course

2. The Decision Lab. `https://thedecisionlab.com/biases`